Lingo Stream

Ambient language learning directly inside YouTube subtitles.

HACKMIT CHINA 2026 | 404 WAIFU NOT FOUND | CHROME EXTENSION · MV3 | V1.0.0

What It Does

A Chrome Extension that injects **micro-immersion** into live YouTube captions — replacing meaningful words with inline translations.

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Original:  I really enjoy learning new skills every day.
Lingo Stream: I really enjoy learning new スキル every day.
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95+	3+	67
LANGUAGES	PROVIDERS	TESTS

Inspiration

Language learners spend hours watching YouTube with subtitles — yet those hours remain **entirely passive**.

We asked: *what if the subtitles themselves could teach you?* Replace a handful of words inline, and the viewer absorbs vocabulary through context — the same mechanism behind natural language acquisition.

Ambient UX meets **spaced-exposure pedagogy** — learning that happens while you watch.

No flashcard decks, no separate apps, no extra effort. Just press play and let your subtitles do the teaching. The result is a frictionless bridge between entertainment and education.

Architecture

A **Caption Observer** watches YouTube's subtitle DOM and feeds each new segment to the **Processor**, which tokenises text, strips stopwords, and picks candidate words at the user's chosen rate.

Candidates hit the **Translation Bridge** — up to five providers with auto-fallback and local caching. Translated tokens replace the originals **inline**, and every swap is saved to **Vocabulary Storage** for quiz review or CSV export.

The **Background Service Worker** coordinates messaging between the content script and the popup UI, managing state persistence across tab navigations and browser restarts via Chrome's storage API.

Key Features

- **Real-time subtitle observation** — Watches YouTube's caption DOM live and reacts to every new segment instantly.
- **Smart stopword filtering** — Skips common words like "the" and "is" so only meaningful vocabulary gets replaced.
- **Configurable replacement %** — Users choose how many words per caption are swapped, from subtle to immersive.
- **Provider auto-fallback** — Cycles through up to five translation APIs so one outage never breaks the flow.
- **Translation caching** — Previously translated words are stored locally, eliminating repeated network calls.
- **Vocabulary + CSV export** — Every learned word is saved and exportable for use with Anki or spreadsheets.
- **Quiz with score tracking** — Built-in active recall quiz reinforces retention and tracks progress over time.
- **50+ target languages** — From Spanish and Japanese to Swahili — learn virtually any language you choose.

Tech Stack

JavaScript · HTML / CSS · Chrome MV3 · Node.js · Vitest · ESLint · Astro · Tailwind · Vite · GitHub CI

Accomplishments

- Shipped v1.0.0 with quiz and vocabulary tracking.
- CI pipeline with Vitest, lint, and coverage on every push to main.
- Modular event-driven architecture with clean layer separation, ready for iteration.
- Full documentation site built with Astro and Tailwind CSS.
- CSV export compatible with Anki and other flashcard tools.

The Science — Why Ambient Learning Works

Linguist Stephen Krashen's **Input Hypothesis** shows that language is acquired when learners encounter *comprehensible input* — material that is mostly understood, with a small stretch of new vocabulary in context. Lingo Stream operationalises this at zero extra effort: the surrounding sentence provides meaning, making each translated word instantly memorable without interrupting the viewing experience.

01 Passive Exposure
Encounter target words inside familiar sentence contexts — the brain anchors new vocabulary to known meaning automatically.

→

02 Spaced Repetition
Words recur naturally across videos. Each re-encounter strengthens the neural pathway without any manual deck management.

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03 Active Recall
The built-in quiz surfaces saved words at the right moment, converting passive recognition into long-term retention.

What's Next — Roadmap

V1.1 · NEAR TERM
Difficulty Adaptive Rate
Auto-adjust the word-swap percentage based on the learner's quiz accuracy — harder for experts, gentler for beginners.

V1.2 · NEAR TERM
Word Hover Tooltips
Hovering over a translated word shows the original, pronunciation guide, and a usage example — all inline without pausing.

V2.0 · MID TERM
Multi-Platform Support
Extend beyond YouTube to Netflix, Coursera, and other captioned streaming platforms using the same caption pipeline.

V2.1 · MID TERM
AI-Personalised Vocab
Use an on-device model to prioritise words at the learner's frontier — maximising acquisition speed per hour watched.

Justin-Yonardo
TEAM LEADER
Primary Engineer — caption pipeline, processor, translation bridge, background worker.

UnoxyRich
UI DESIGN & OPS
UI Design, Video Editing, Communication Materials, GitHub Actions & org setup.

LeoLiu32
UI IMPLEMENTATION
User Interaction, UI Coding, popup interface, quiz flow & frontend polish.

Lingo Stream
Learn languages while you watch.
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